Flood Relief Management System

CSC 1051 – Full Stack Fundamentals Laboratory

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GitHub repository link : https://github.com/binuri2003/FLOOD-RELIEF-MANAGEMENT-SYSTEM.git

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1.INTRODUCTION

1.1 DESCRIPTION ABOUT PROJECT

Floods are one of the most common natural disasters in Sri Lanka. During flood situations, it is difficult to collect accurate information about affected people and their relief needs. Manual data collection is slow and error prone. Therefore, a computerized system is required to manage flood relief information efficiently.

The flood relief management system is a web-dased system developed to collect and manage data related to flood-affected people. The system allows affected individuals to register, log in and submit relief requests. Administrators can manage users and view summarized reports to support decision-making.

1.2 OBJECTIVES

- To provide an online platform for flood-affected people or relief people
- To manage user information securely using authentication
- To allow administrators to monitor flood relief data
- To generate summarized reports for better decision making
- To reduce manual work and improve data accuracy

2.SYSTEM OVERVIEW

2.1 OVERALL SYSTEM DESCRIPTION

The Flood Relief Management System is a web-based application developed to support the effective management of flood relief activities through a centralized digital platform. The system is designed to assist affected persons in submitting and managing relief requests while enabling administrators to monitor, manage, and report on system data efficiently.

The system provides secure user authentication with role-based access for affected persons and administrators. New users can register by filling in the required registration details, after which they can log in and access system features based on their assigned role. User information is stored and maintained by the system to ensure proper access control and data management.

Affected persons can submit flood relief requests by completing a relief request form that captures all necessary information related to their situation and needs. Once submitted, users can view their own request history, update request details, or delete requests when required. This functionality ensures transparency and allows users to manage their requests conveniently.

Administrators are responsible for managing the overall system. They can view all registered users and access detailed user information presented in a structured format. The system also provides reporting and filtering features that allow administrators to generate summarized system data based on criteria such as area and type of relief request. These summaries are automatically updated as system records change, supporting effective monitoring and decision-making. Administrators also have the authority to delete user accounts when necessary.

Overall, the Flood Relief Management System enhances efficiency, accuracy, and transparency in flood relief operations by reducing manual processes and providing a reliable platform for managing users, relief requests, and administrative reporting.

2.2 TARGET USERS

- Affected Persons: - Individuals needing food, medicine,shelters etc
- Administrators: - Government or NGO officials monitoring relief distribution

2.3 TECHNOLOGY USED AND HOW YOU INTEGRATED THEM

Version control and collaboration for the system development were managed using GitHub. A central repository was created for the project, and separate branches were assigned to each group member. The overall system was divided into sub-parts, and each member developed their assigned component within their own branch. After completing individual tasks, changes were pushed to GitHub and merged into the main branch to form a fully functional system. Final adjustments and testing were carried out after merging all branches.

For the frontend development, HTML was used to structure web pages, while CSS was applied for styling and layout design. JavaScript was used to add interactivity and display pop-up messages to enhance user experience. Additionally, the Bootstrap framework was used on the frontend to create responsive and visually consistent interfaces.

The backend of the system was developed using PHP, which handles server-side logic such as user authentication, relief request processing, and communication with the database. In accordance with the project requirements, no backend frameworks were used, and all functionality was implemented using core PHP.

For data storage and management, the system uses a MySQL database, which stores user information, relief requests, and system data. The database was managed using phpMyAdmin, and the system was deployed in a local development environment using the XAMPP server, which integrates Apache, PHP, and MySQL to enable smooth interaction between the frontend, backend, and database components.

2.4 DATABASE DESIGN

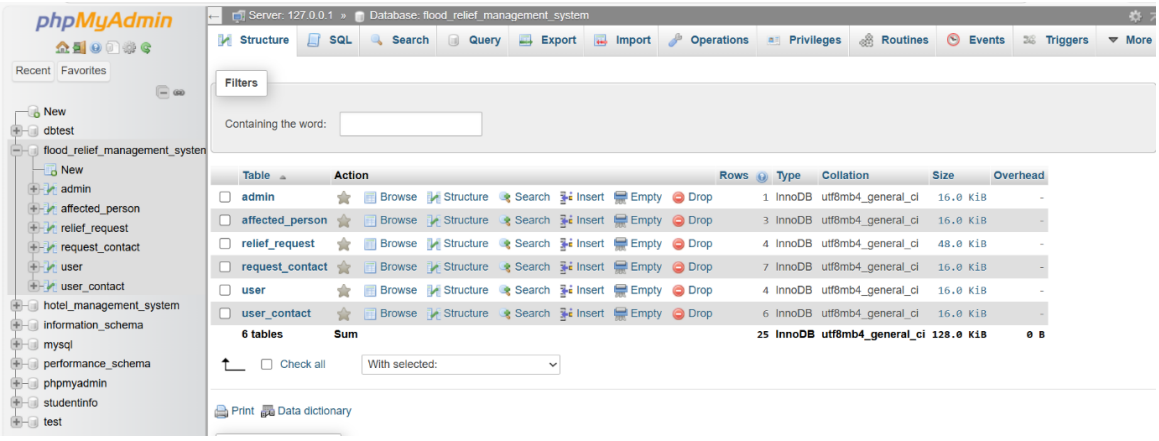


Figure 1 DataBase Design

In designing the Flood Relief Management System database, we first created an Enhanced Entity Relationship (EER) Diagram to model the system's entities, attributes, and relationships. Based on the EER diagram, we derived the relationship schema and created the corresponding database tables.

The database includes the following main entities:

- user
- affected_person
- admin
- relief_request

Multi valued attributes such as contacts are separated into their own tables (request_contact and user_contact) to maintain normalization.

Database Tables

1. **user Table**

    - Primary Key: UserID (AUTO_INCREMENT)
    - Stores general information of all users.
    - Acts as the parent table for both admin and affected_person.
    - Contains a Role attribute to identify user type.

2. **admin Table**

    - Primary Key: UserID
    - Foreign Key: UserID from user table and Uses the same UserID as in the user table (not auto-generated).
    - Stores information about system administrators.

3. **Affected_person Table**

    - Primary Key**:** UserID from user table
    - Foreign Key**:** UserID
    - Stores details of individuals affected by floods.

4. **Relief Request Table**

    - Primary Key**:** RequestID (AUTO_INCREMENT)
    - Stores information about requests for relief by affected persons.
    - Foreign Keys:
        AffectedUserID → references affected_person(UserID)
        AdminUserID → references admin(UserID)
    - Contains details such as location, severity level, and description.

5. **request_contact Table**

    - Primary Key**:** RequestID + ContactNumber
    - Foreign Key**:** RequestID from relief_request table
    - Stores multiple contacts for each relief request.

6. **User_contact Table**
    - Primary Key**:** UserID + ContactNumber
    - Foreign Key**:** UserID from user table
    - Stores multiple contacts for each user account.

Key Design Points

- Auto-increment is used only for user and relief_request tables to ensure unique identification.
- The admin and affected_person tables reuse UserID from the user table to represent specialization (inheritance).
- Foreign keys are used to maintain referential integrity between related tables.
- ON DELETE CASCADE ensures that when a parent record is deleted, related child records are also automatically removed.
- Multi-valued attributes (contact numbers) are separated into dedicated tables to achieve normalization and reduce redundancy.
- Each relief request can be linked to one affected person and optionally handled by an admin.
- Each user and each relief request can have multiple contact numbers.

This structure supports efficient management of users, affected persons, and relief requests while maintaining data consistency and scalability.

2.5 BACKEND LOGIC

2.5.1 USER AUTHENTICATION

Registration Logic

- Users submit the registration form with details such as first name, last name, address, username, and password.
- All fields are validated to ensure no required information is missing.
- All new users are registered as normal users by default.
- Registration does not grant any administrative privileges. Users can only log in as normal users until their role is updated by the system owner.

Login Logic

- When a user enters their username and password, the system checks the database for matching credentials.
- If the username does not exist, the system blocks access and shows an alert asking the user to register first.
- If the username exists but the password is incorrect, access is denied and an error message is displayed.
- If the credentials are valid, the system retrieves the user's role from the database and redirects them accordingly.

Role-Based Access Control

- After successful login, the system checks the user's role,

  Normal User → redirected to the user dashboard

  Admin → redirected to the admin control panel

- Role checks are done on the backend to prevent unauthorized users from accessing restricted pages.

Admin Controller Role Management

- Role assignments are controlled exclusively by the system owner through the admin controller panel.
- The system owner can select registered users & change their role from user to admin.
- Once a user is assigned to the admin role, they can log in using the same login interface and access the admin panel. Also, Normal users cannot modify roles or access administrative functionalities

Logout Logic

- When a user logs out, the system destroys the active session.
- The user is redirected to the login page.
- This prevents unauthorized access after logout.

2.5.2 RELIEF REQUEST MANAGEMENT

The Relief Request Management module of the Flood Relief Management System is designed to enable affected individuals to submit and manage relief requests efficiently. After logging into the system, users are directed to a user home interface that contains a navigation bar with options to create a new relief request or view their request history. This interface functions as a navigation layer only and does not include backend processing. The purpose of this design is to provide users with a simple and clear workflow for accessing system functionalities.

When a user selects the option to make a relief request, they are redirected to a request form where detailed information is collected. The backend of this form is implemented using the PHP programming language, which handles data processing and database interactions. The form is structured into several logical sections to ensure organized data collection. User information fields include the User ID and username, which are retrieved as part of the user's registered details. Household related information includes the contact person's name, contact numbers, residential address, and the number of family members affected.

In addition to user and household information, the form also captures relief and location details required for effective flood relief management. These fields include the type of relief requested, district, divisional secretariat, Grama Niladhari (GN) division, flood level, and a brief description of the situation. Once the form is submitted, the PHP backend validates and stores the data in the relief_request table in flood_relief_management_system database. Each request is assigned a unique, automatically generated request ID to ensure accurate identification and record management.

The request history feature allows users to view all relief requests they have submitted, filtered according to their UserID. This ensures that users can only access their own request records. The history section also provides options to edit or delete existing requests. When a request is deleted, the corresponding record is permanently removed from the database. When a request is edited, the updated information replaces the previous data while maintaining the same RequestID. Overall, this module ensures systematic data collection and efficient request management for affected individuals during flood situations.

2.5.3 USER MANAGEMENT & REPORTING

The admin backend is developed using PHP and MySQL to manage user data, reporting, and deletion while maintaining database integrity. It acts as the central controller by handling frontend requests and returning responses in JSON format.

When the admin panel loads, the backend processes GET requests to fetch all users, individual user details, or summary reports. User data such as User ID, Name, Username, Address, and Role are retrieved from the user table. Individual user details are displayed dynamically without reloading the page.

Summary reports are generated using SQL queries with filters like district and relief type. The system calculates:

- Total registered users
- High severity relief requests
- Requests based on selected relief type

These results are sent as JSON, enabling real-time updates on the dashboard.

Deletion Handling

User deletion follows a hybrid cascading approach:

- Automatic cascading (ON DELETE CASCADE) removes related records in user_contact, admin, affected_person, and request_contact
- Manual deletion is required for relief_request before deleting a user

Deletion steps:

1. Delete related relief requests
2. Cascading removes related contacts and role records
3. Delete the user

This ensures no orphan data and avoids foreign key errors.

Security

Basic input validation and structured SQL queries are used. JSON responses ensure smooth frontend backend communication. The backend supports efficient and dynamic system operations.

2.6 FRONT END FUNCTIONALITY

The frontend is developed using HTML, CSS and JavaScript. It provides simple and user-friendly interfaces for users and administrators. JavaScript is used to show confirmation pop-up messages and improve users' interaction.

2.6.1 USER AUTHENTICATION

- The system provides registration and login forms for users.
- Allows users to enter email and password to access the system
- Verifies user credentials through the backend
- Supports role-based access
- Redirects users to correct dashboard after login
- Provides logout functionality to end the session securely
- Displays alert or confirmation messages for login, logout and errors

2.6.2 RELIEF REQUEST MANAGEMENT

- The system provides a user-friendly front-end interface designed for affected individuals in the Flood Relief Management System.
- A navigation bar is displayed on the user interface, allowing users to navigate between key system functions such as Make Request, View History, and Logout.
- The Make Request functionality includes a structured relief request form that enables users to enter all required information and submit a new relief request.
- The relief request form contains input fields for entering details and includes a dropdown menu for selecting the type of relief and radio buttons for selecting the flood level, ensuring accurate and standardized data input.
- A Submit button is provided to allow users to send the completed form to the system.
- The system performs form validation on the front end, and if required fields are missing or invalid data is entered, appropriate error messages are displayed to the user.
- The homepage includes a footer section displaying the organization's contact details, enabling users to easily reach the responsible authorities.
- The View History functionality allows users to retrieve their previously submitted relief requests by entering their user ID and clicking the Load History button.
- Each relief request displayed in the history section includes Edit and Delete buttons for managing submitted requests.

- When the Edit button is selected, users can update the details of an existing relief request while keeping the same request ID.
- When the Delete button is selected, a confirmation pop up message is displayed asking the user to confirm whether they really want to delete the selected record.
- The relief request is deleted only after the user confirms the deletion action through the pop-up message.
- After successful actions such as request creation, update, or deletion, the system displays success pop up messages to notify the user.
- These confirmation and error pop-up messages improve system usability by preventing accidental actions and clearly informing users about errors and successful operations.

2.6.3 USER MANAGEMENT & REPORTING

- Provides an admin dashboard interface for system management.
- Displays all registered users in a structured table format on the admin panel.
- Shows user details such as User ID, Full Name, Username, Address, and Role.
- Allows the admin to select a user and view detailed user information.
- Allows the admin to delete registered users when required.
- Shows confirmation pop-up messages before deleting users to prevent accidental removal.
- Provides a district selection dropdown that allows the administrator to filter system reports based on areas.
- Provides a relief type selection option to filter reports based on the type of relief requested (Food, Water, Medicine, Shelter).
- Displays system summary reports for the administrator.
- Reports include Total registered users, High severity households And Number of requests based on relief type
- Report data updates dynamically without reloading the page whenever the filter criteria change.

3. RESULT

3.1 SYSTEM OUTPUTS

The Flood Relief Management System produces several outputs based on the actions performed by users and administrators. These outputs are displayed through web interfaces, confirmation messages, and system reports. They help users understand the results of their actions and allow administrators to monitor and manage flood relief information effectively.

The system outputs are mainly generated through three functional areas user authentication, relief request management, and administrative management and reporting.

3.1.1 USER AUTHENTIFICATION OUTPUT

The user authentication module generates outputs related to user registration, login, and logout operations.

When a new user submits the registration form, the system validates the entered data and stores the user details in the database. After successful registration, a confirmation message is displayed indicating that the account has been created successfully. This confirms that the user can now log in using the registered credentials.

During login, the system checks the entered username and password against the records stored in the database. If the credentials match, the user is redirected to the appropriate dashboard based on their assigned role.

Normal users (affected persons) are redirected to the user dashboard where they can submit and manage relief requests, while administrators are redirected to the admin dashboard where management and reporting functions are available.

If the username does not exist, the system displays an alert asking the user to register before logging in. If the password is incorrect, an error message is shown indicating invalid login credentials. These outputs help maintain secure access control. When a user logs out, the system terminates the active session and redirects the user back to the login page, preventing unauthorized access.

3.1.2 RELIEF REQUEST OUTPUT

The relief request management module generates output when affected individuals create, view, update, or delete relief requests.

When a logged-in user submits a relief request through the request form, the system processes the information and stores it in the database. Each request is assigned a unique request ID. After successful submission, the system displays a pop-up confirmation message indicating that the request has been created successfully.

Users can access the View History section to see them previously submitted requests. The system displays these requests in a structured format, allowing users to track and review their information.

Users can also edit existing requests. When the edit option is selected, the system loads the stored information into the form so the user can modify it. After submission, the updated data replaces the previous record, and a confirmation message is displayed.

Users can delete their own requests when necessary. When the delete option is selected, the system displays a confirmation pop-up message. Once confirmed, the system removes the request from the database and displays a successful message.

3.1.3 ADMIN SYSTEM OUTPUT

The administrator interface generates output related to user management and system reporting.

The admin dashboard provides an overview of the system and allows administrators to access management functions. One of the main outputs is the registered user list, displayed in a table format with information such as user ID, full name, username, address, and assigned role.

Administrators can select a user from the table to view more detailed information. This helps monitor user accounts and verify system records.

The system also generates summary reports that provide an overview of flood relief requests submitted by affected individuals. These reports include the total number of registered users, the number of high severity households, and the number of relief requests categorized by relief type such as food, water, medicine, or shelter.

The admin panel provides filtering options such as district and relief type. When the administrator selects a filter, the system retrieves relevant data from the database and updates the summary values.

These report results are displayed dynamically on the dashboard without refreshing the page, allowing administrators to monitor relief data efficiently.

Overall, the system provides clear feedback to users and helps administrators monitor flood relief activities effectively.

3.2 SCREENSHOTS OF THE SYSTEM COMPONENTS

3.2.1 USER AUTHENTICATION

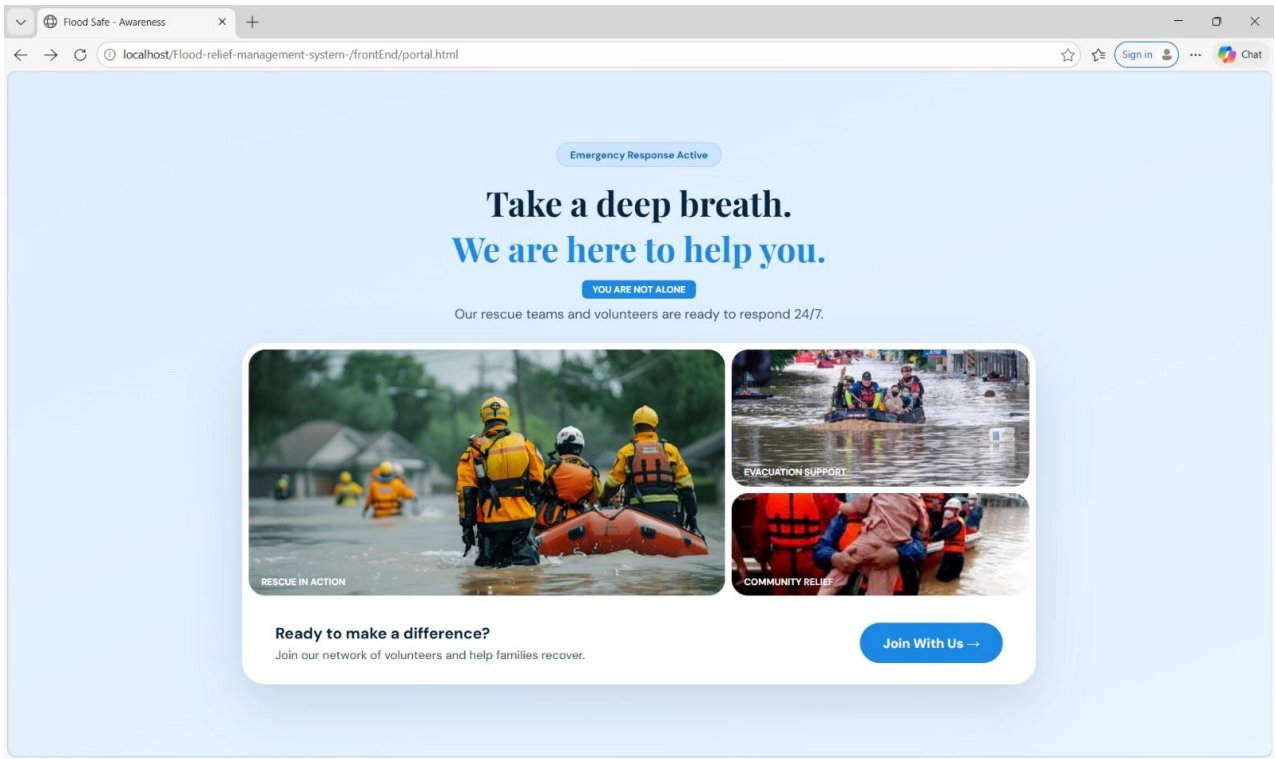


Figure 1(a) System Main FrontEnd

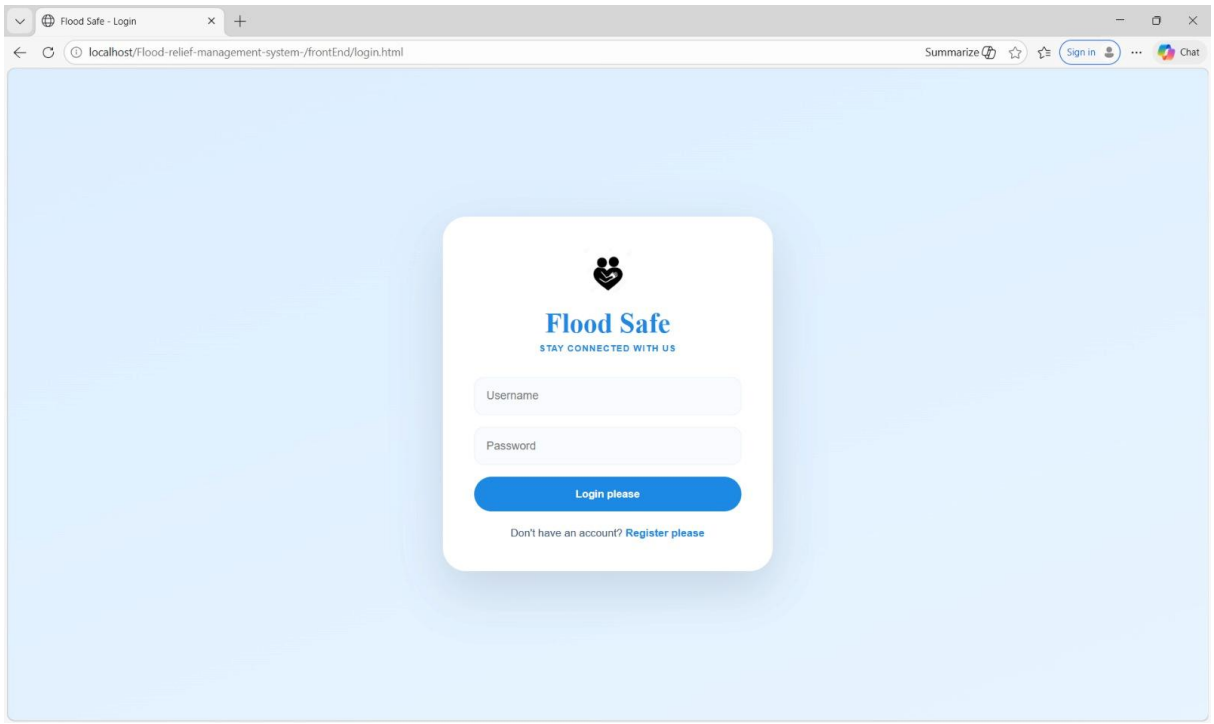


Figure 1(b) Login Interface

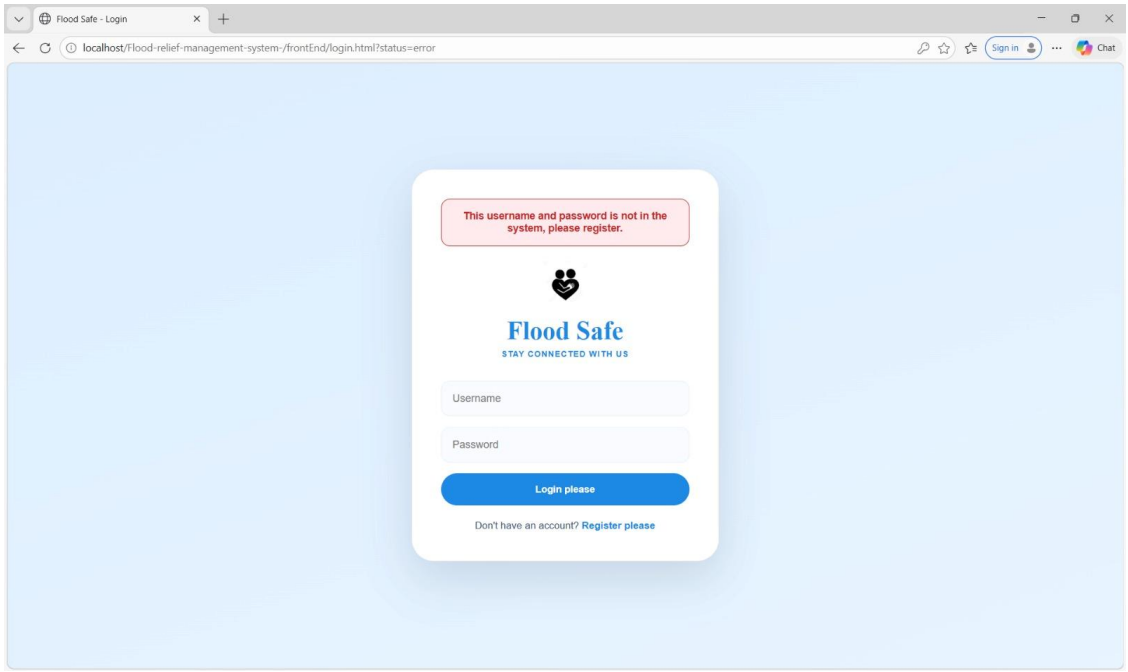


Figure 2(a) invalid Login Interface

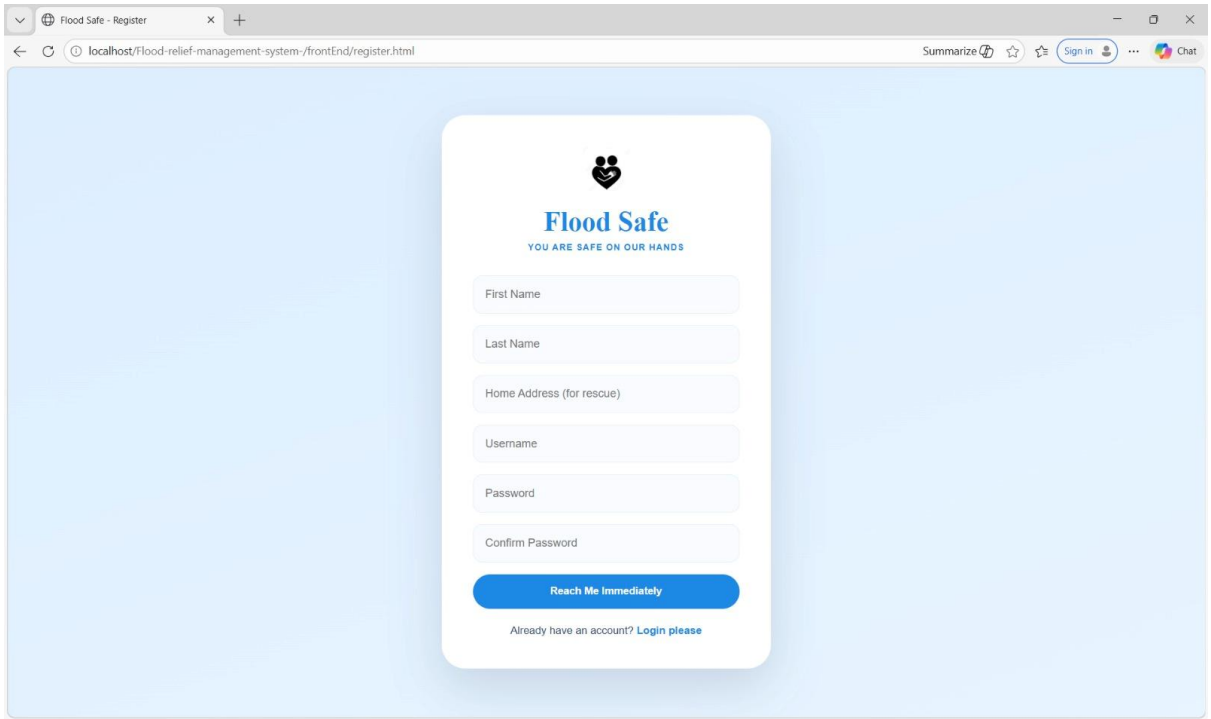


Figure 3(b) Registration Interface

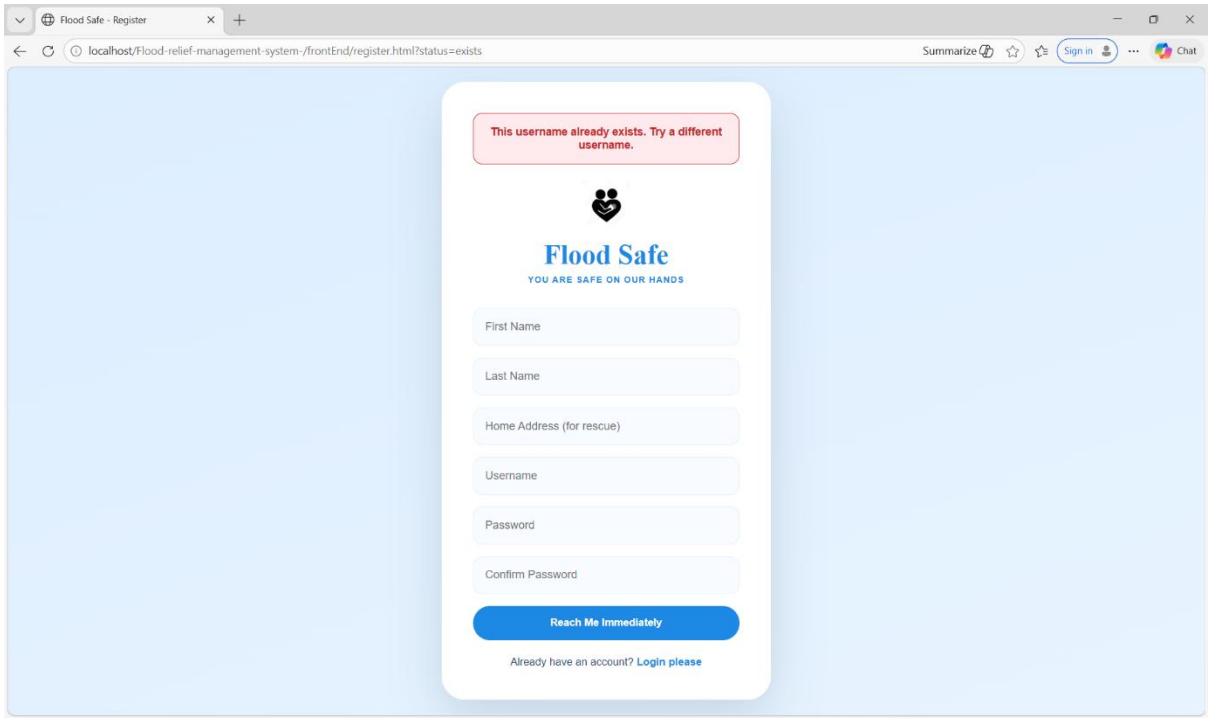


Figure 4(c) Invalid Registration

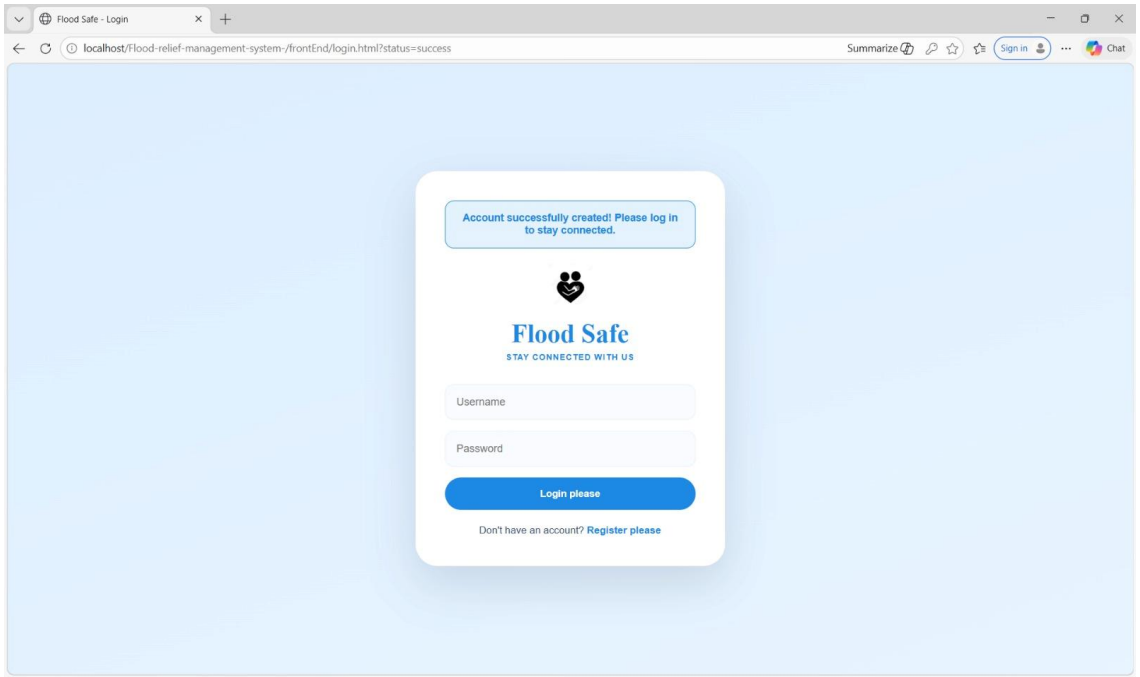


Figure3(a) Registration Successful Notification

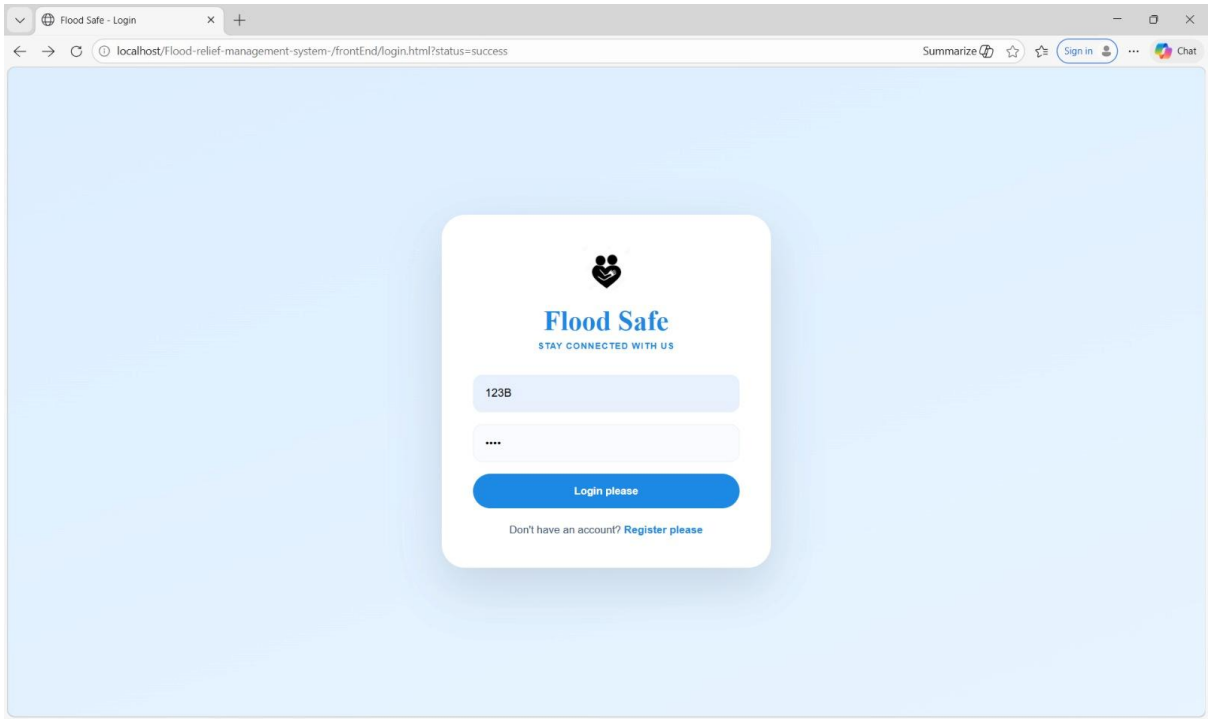


Figure3(b) Sample Login Process

3.2.2 RELIEF REQUEST MANAGEMENT

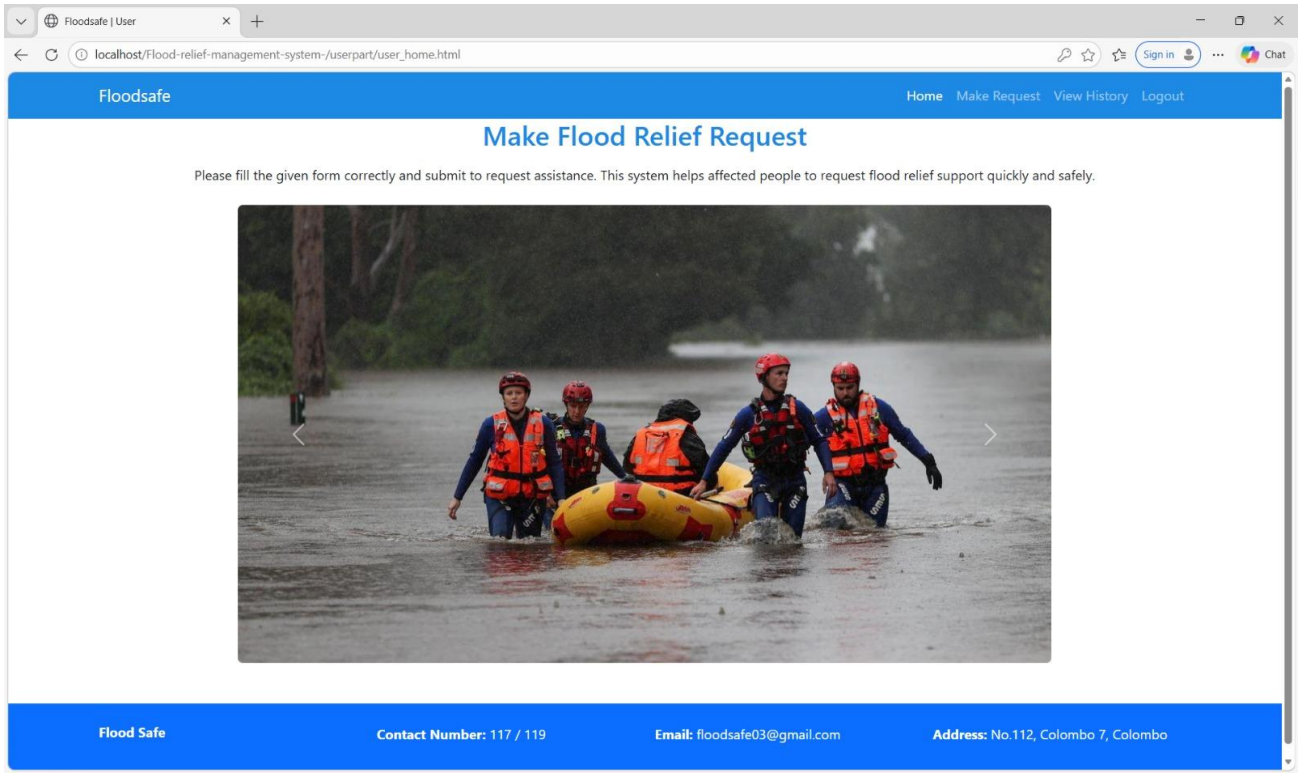
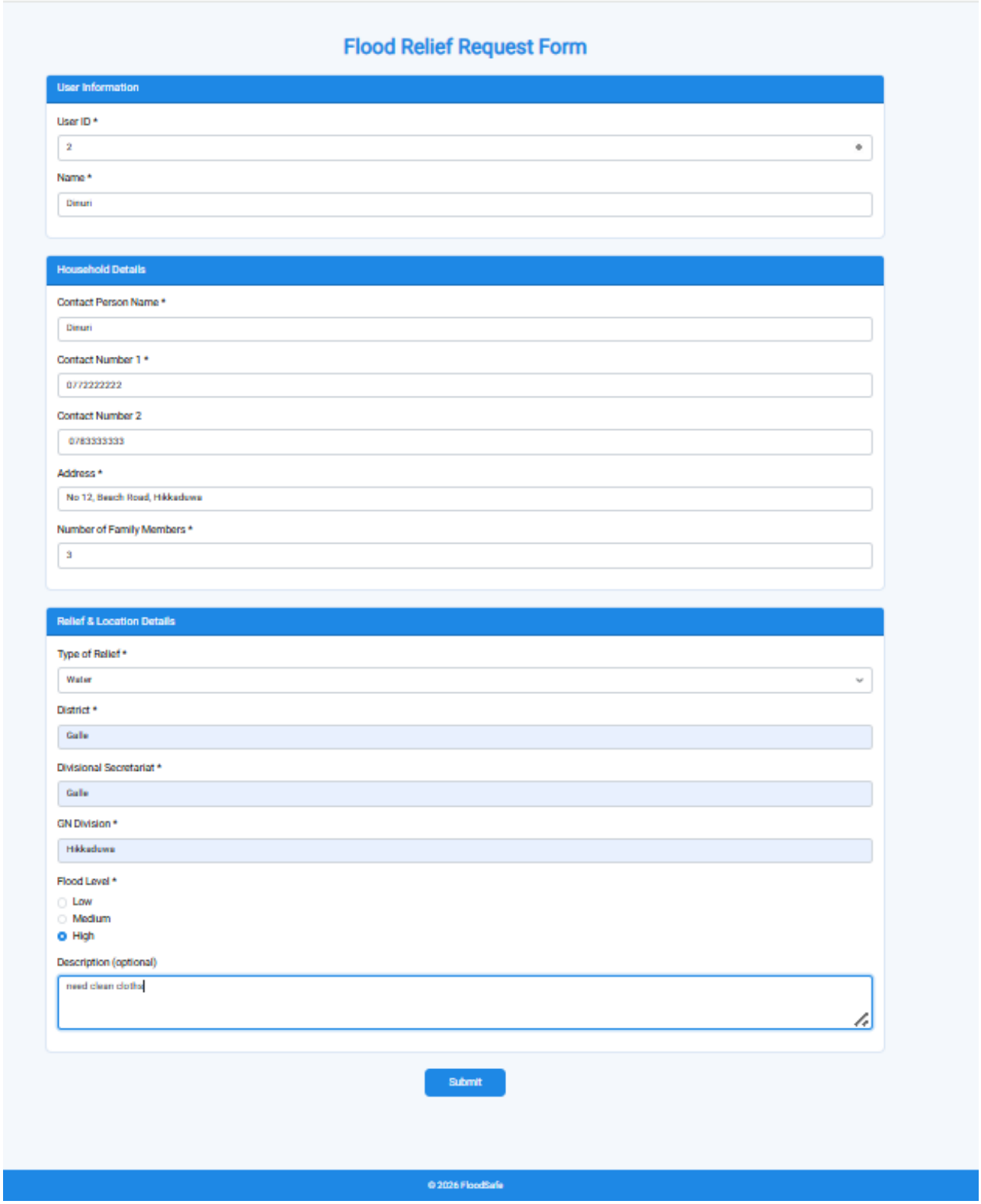


Figure 4 User Home Page



Flood Relief Request Form

User Information

User ID *

2

Name *

Dinuri

Household Details

Contact Person Name *

Dinuri

Contact Number 1 *

0772222222

Contact Number 2

0783333333

Address *

No 12, Beach Road, Hikkaduwa

Number of Family Members *

3

Relief & Location Details

Type of Relief *

Water

District *

Galle

Divisional Secretariat *

Galle

GN Division *

Hikkaduwa

Flood Level *

- Low
- Medium
- High

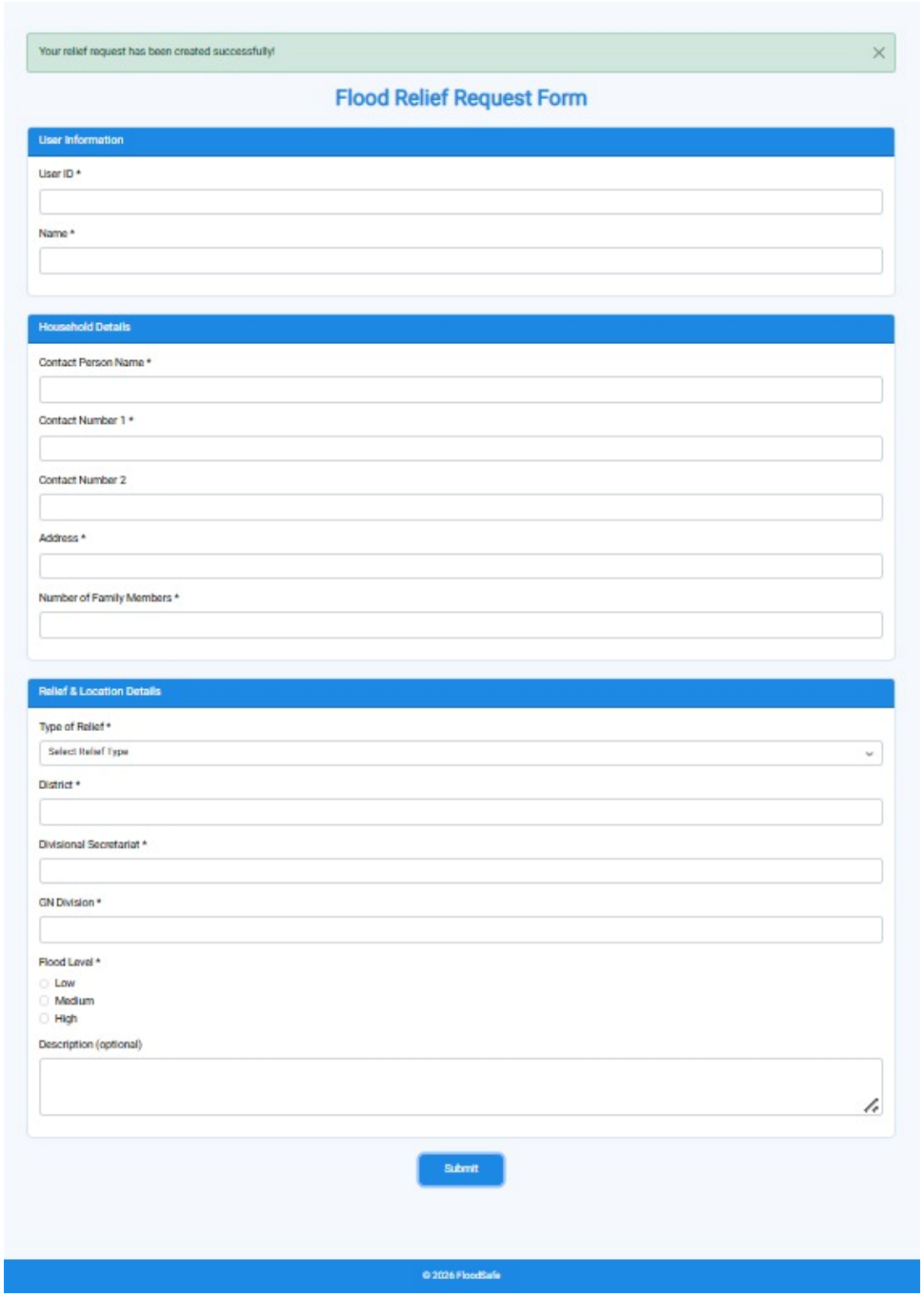
Description (optional)

need clean clothe

Submit

© 2026 FloodSafe

Figure 5 Request Form



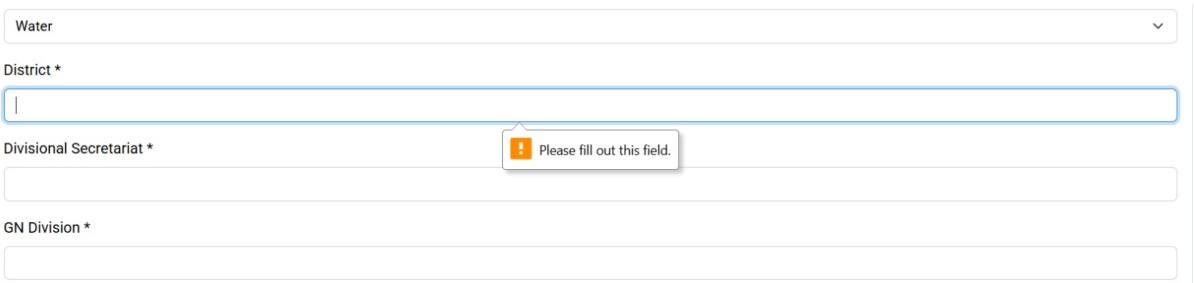


Figure 6 Request Created Successfully (pop up message) and please fill required messages pop up message

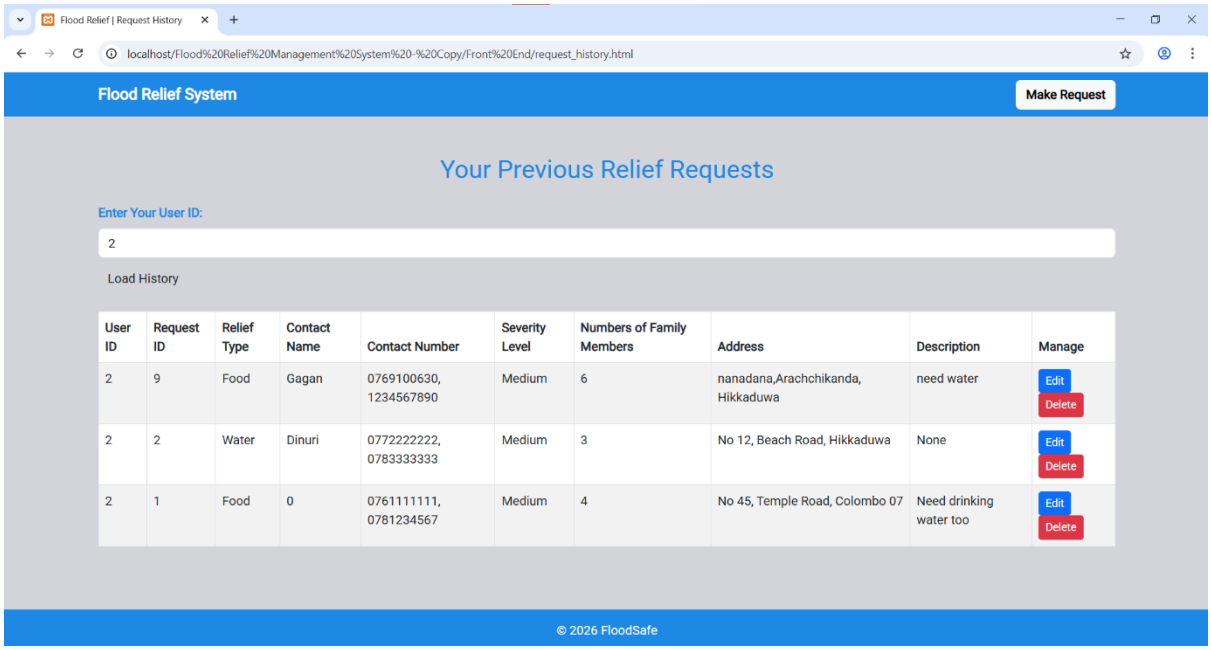


Figure 7 View Own History

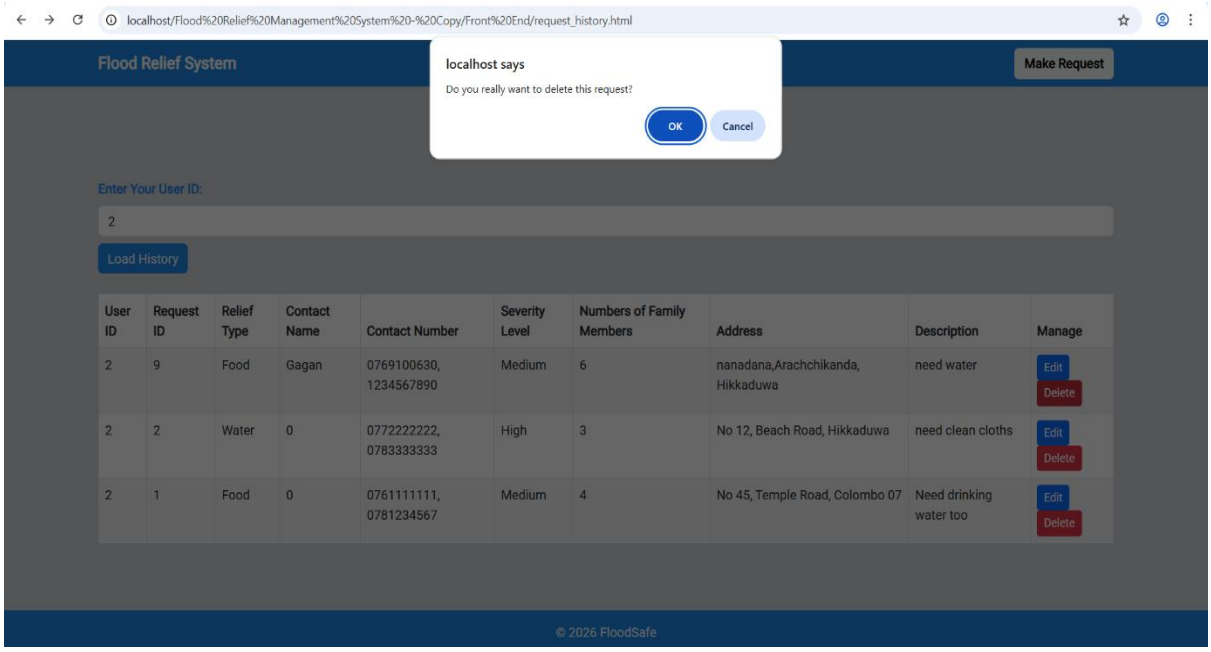


Figure 8 Confrirm Message For Deleting Process

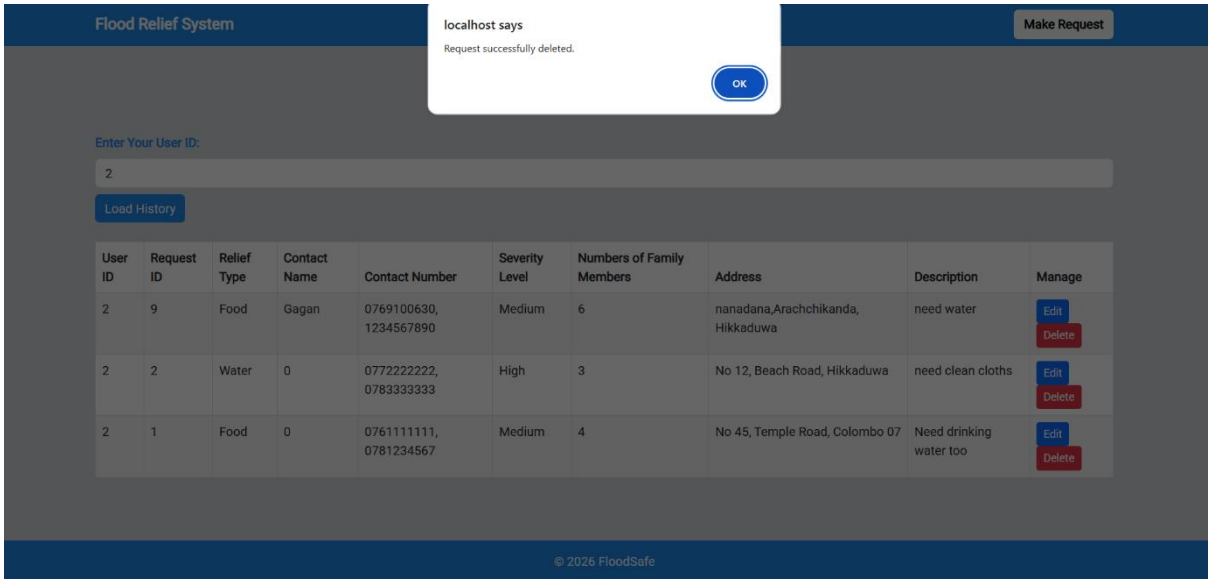


Figure 9 Successfully Deleted

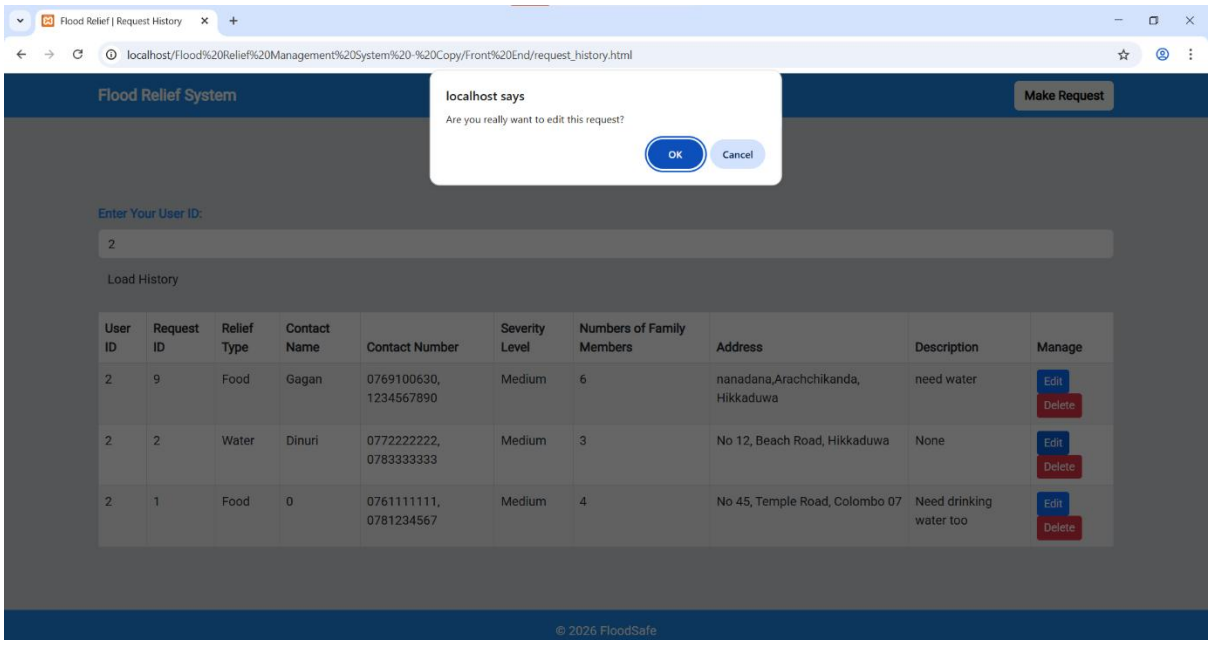


Figure 10 Confirming Message For Editing Process

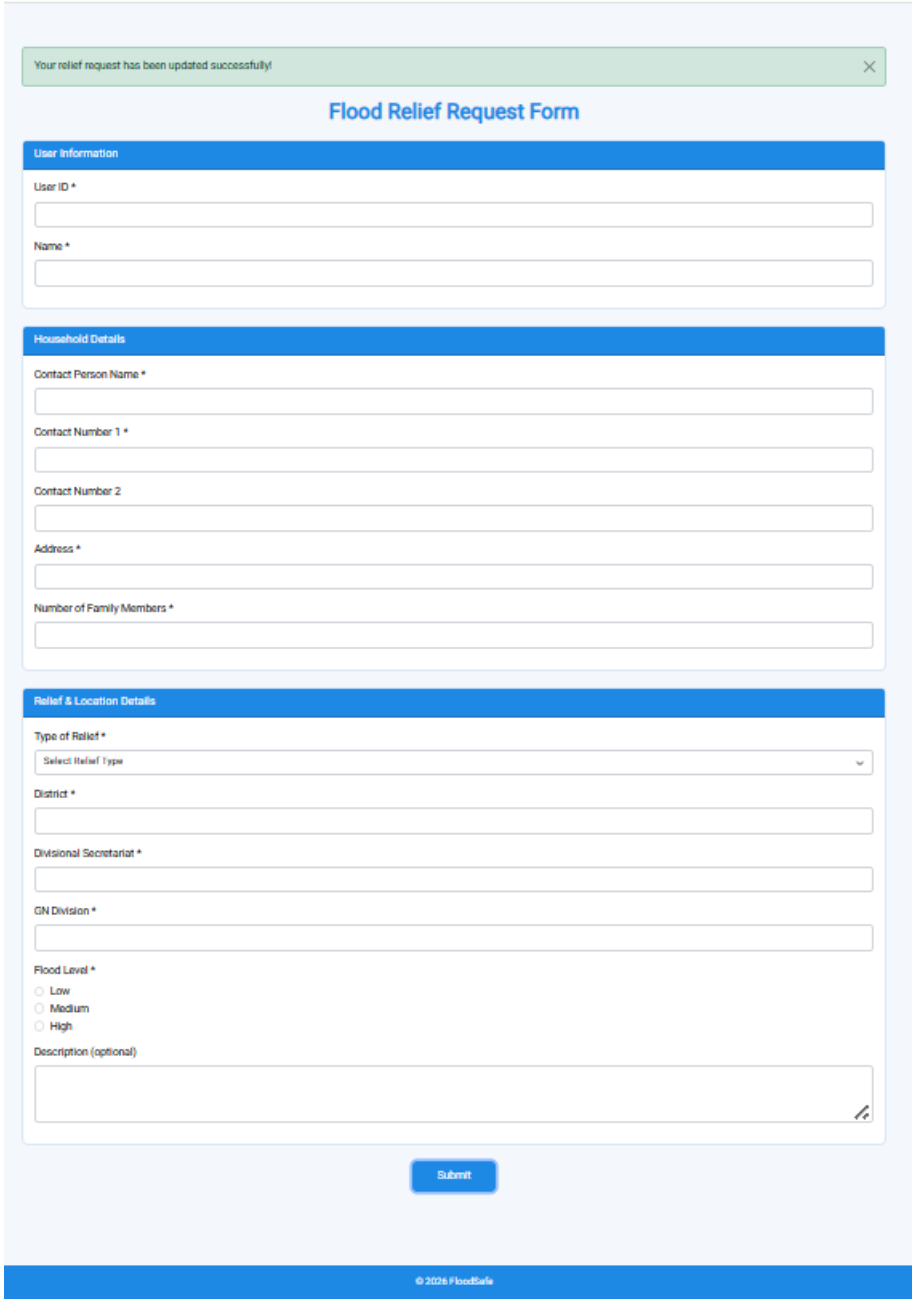


Figure 11 Successfully Edited

3.2.3 USER MANAGEMENT & REPORTING

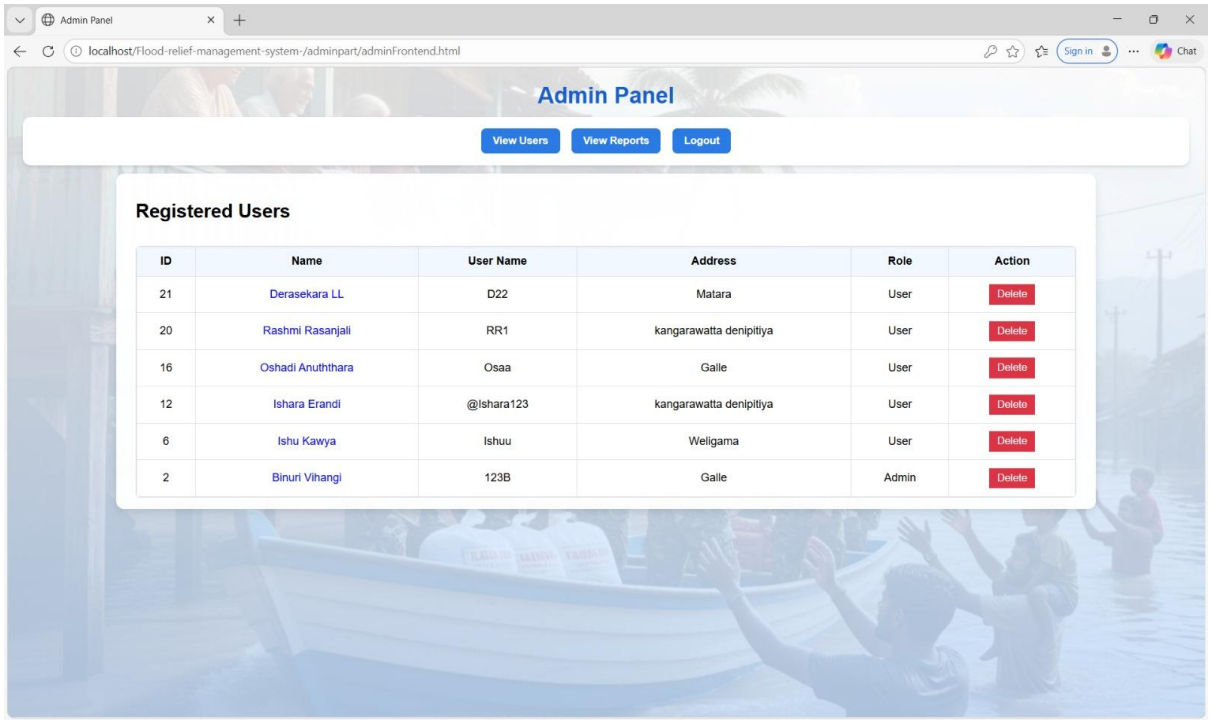


Figure 12(a) admin panel Interface

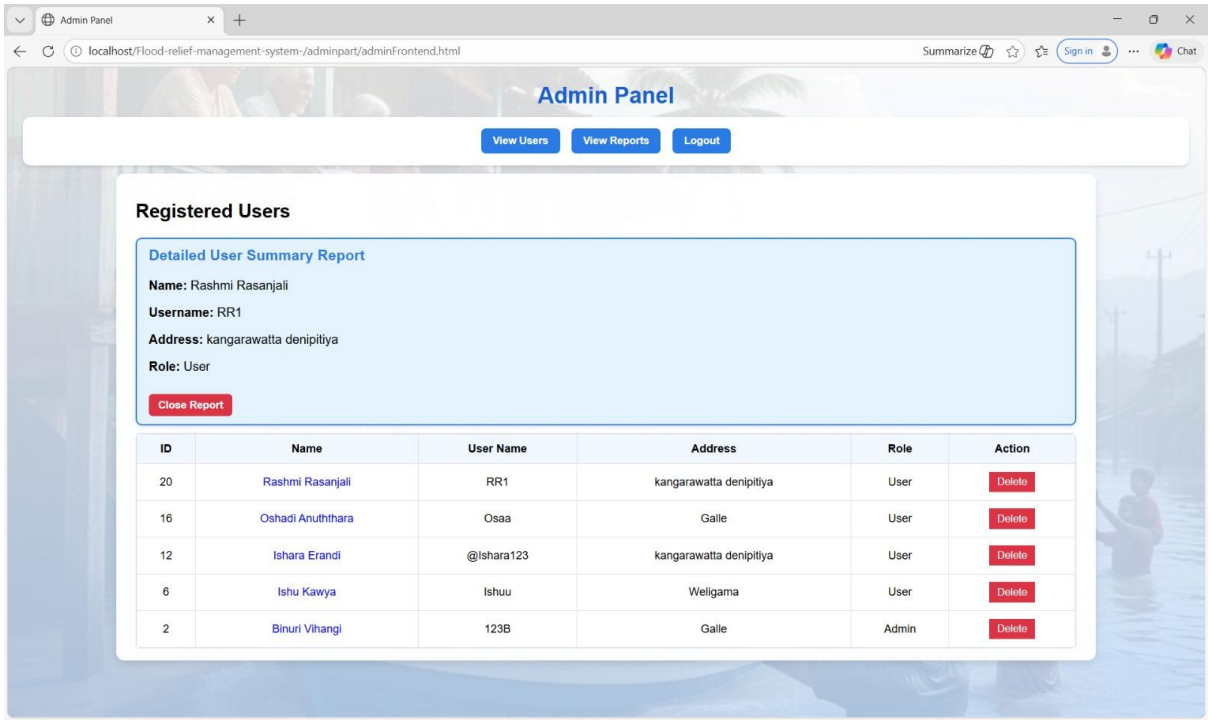


Figure 12(b) Detailed Summary Report

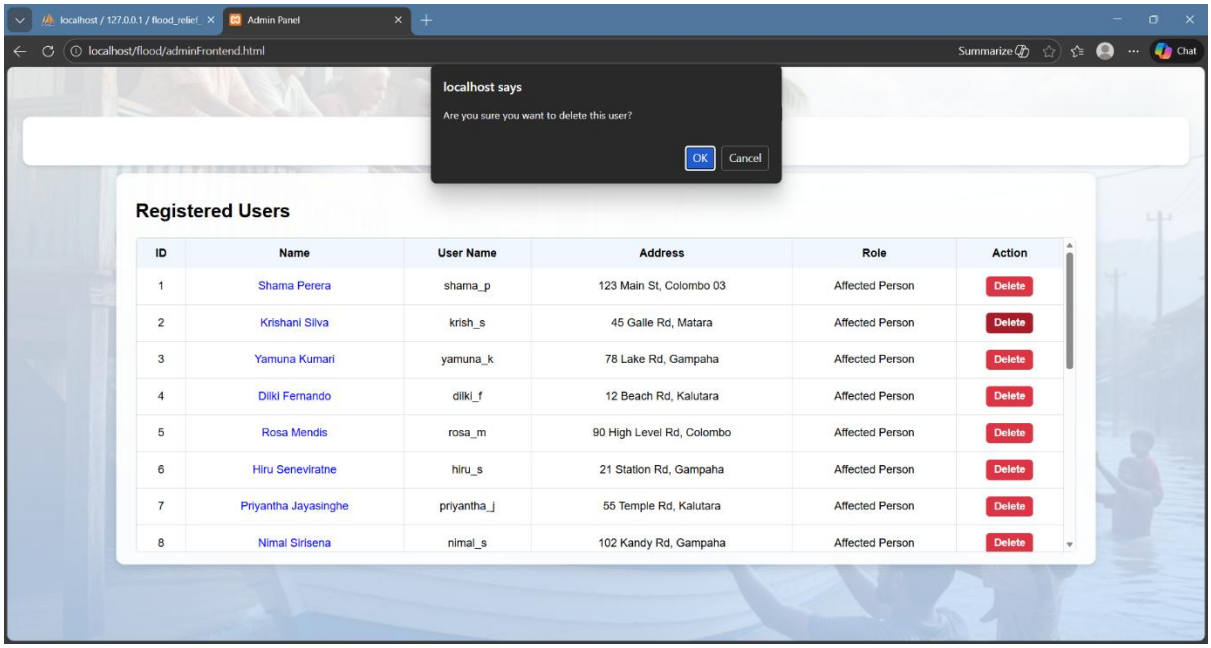


Figure 12(c) Pop up Message for delete confirmation

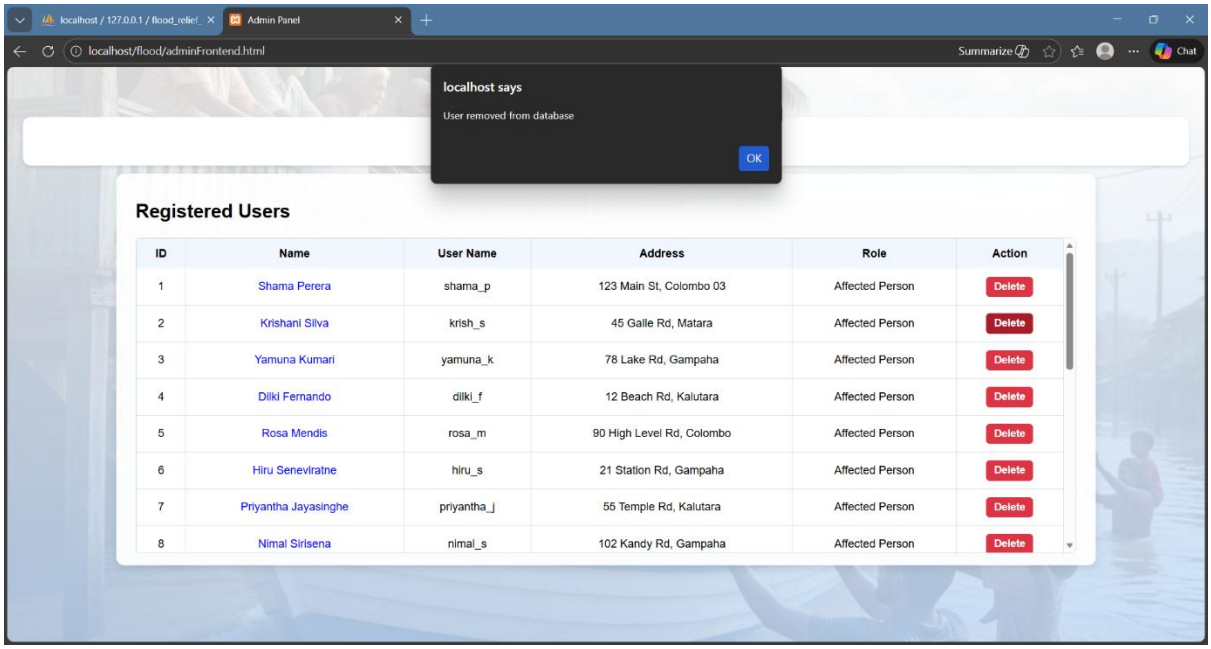


Figure 12(d) Delete from the databse pop up Message

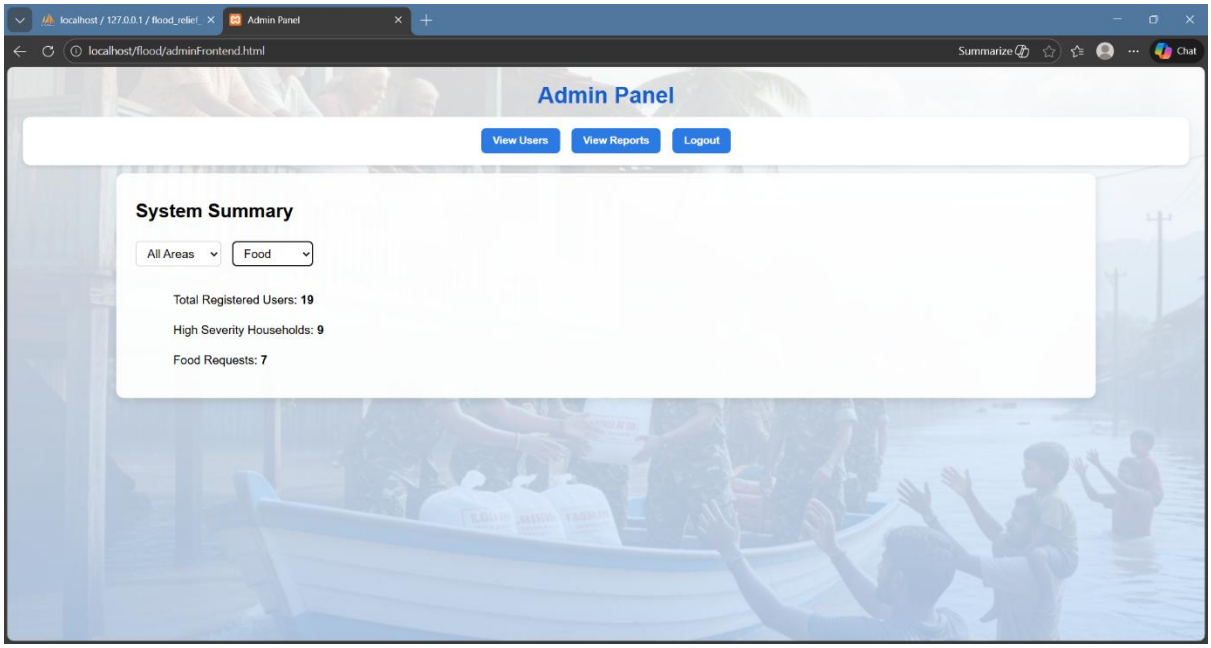


Figure12(e) View Report Section

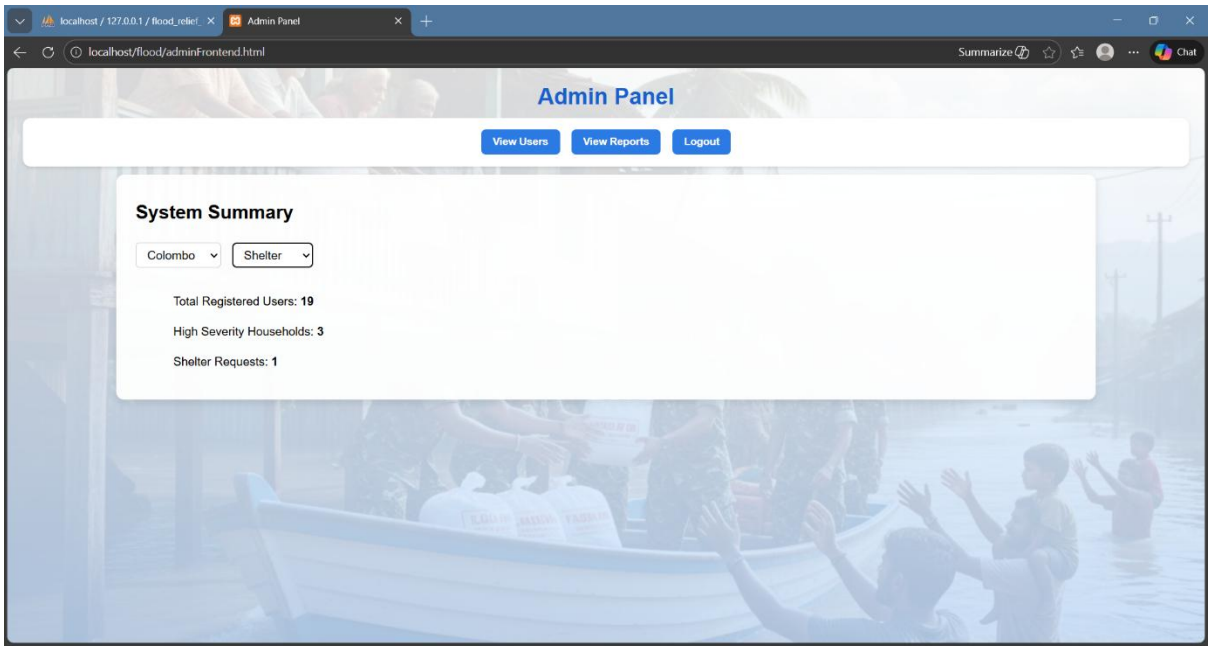


Figure12(f) Summary report by filtering

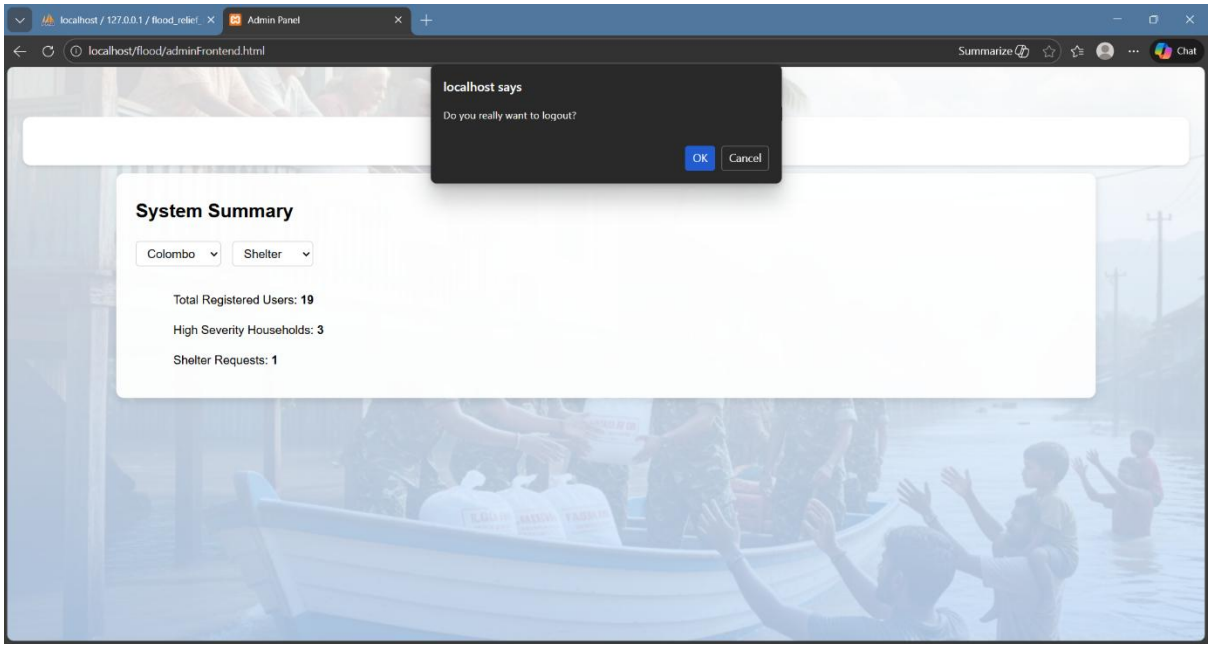


Figure 12(g) Logout confirm pop up message

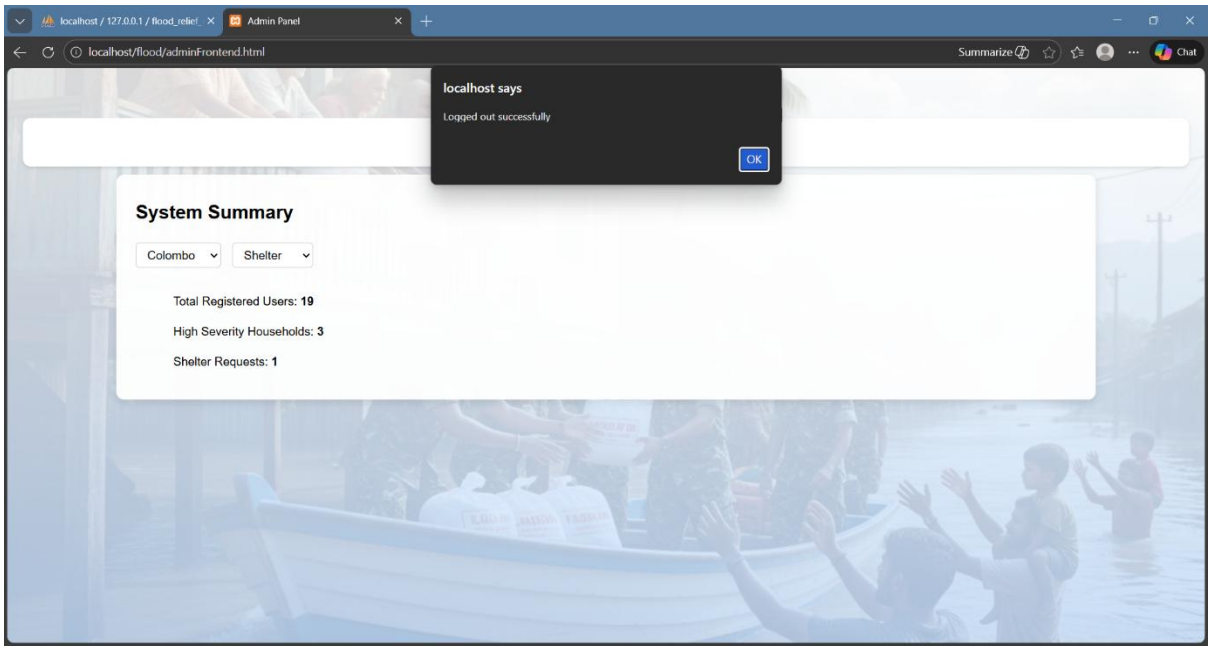


Figure 12(h) Logged out successfully pop up message

3.3 SUMMARY OF THE SYSTEM

The Flood Relief Management System was successfully developed as a full-stack web application using HTML, CSS, JavaScript, PHP, and MySQL. The system provides an organized digital platform to manage flood relief information efficiently.

The application supports two main user roles: affected persons and administrators. Affected people can register, log in, and submit flood relief requests containing important household and location details. They are also able to manage their requests by viewing, updating, or deleting them when necessary.

Administrators have access to a management dashboard that allows them to monitor the system, view all registered users, and generate summarized reports based on relief data. The reporting feature enables administrators to analyze system activity through dynamically generated summaries.

The system integrates frontend interfaces with backend logic and database operations to ensure smooth data flow and proper system functionality. Validation mechanisms and confirmation messages improve usability while preventing incorrect operations.

Overall, the system demonstrates how full-stack technologies can be used to create an effective information management platform that supports flood relief operations, improves data accuracy, and reduces manual work during disaster situations.

4. INDIVIDUAL CONTRIBUTION

Name:-M.A.B.Vihangi

Student ID:-AS20240449

I contributed to the Flood Relief Management System by designing the database structure and implementing key system functionalities. My work covered the full development process, starting from EER modeling to database creation and full stack web development.

My Contributions

- Designed EER (Enhanced Entity Relationship) diagrams for the Flood Relief Management System.
- Converted the EER diagrams into a relational schema.
- Created and implemented the SQL database using the relational schema.
- Developed the front end using HTML, CSS, and JavaScript.
- Developed the back end using PHP.
- Implemented Relief Request Management (Affected Person role), including:
- Home page
- Relief request submission form
- View own request history
- Edit and delete relief request functions
- Make pop up messages.
- Provided support for final report preparation and documentation.

This contribution ensured a well structured database and a functional relief request management process. The implemented features support effective interaction for affected persons while maintaining proper system integration.

Signature :

Student Name : M.P.G.D.Nimeshika

Student ID : AS20240479

I was responsible for developing the Admin Frontend part of the Flood Relief Management System. My main task was to design and implement the admin user interface using the web technologies taught in this course.

My contributions :

- Designing the Admin Panel interface using HTML
- Styling the admin pages using CSS to make them clear and user-friendly
- Using JavaScript to switch between different sections of the page
- Implementing confirmation pop-up messages for actions such as deleting users
- Organizing admin frontend files into a separate folder for better structure
- Provided support for final report preparation and documentation.

Through this work, I gained a better understanding of frontend development and learned how HTML, CSS, and JavaScript work together in a full-stack web application.

Signature :

Name – Roshanki Savindi
Index Number – AS20240387

My Contribution – Admin Backend, JavaScript & Database Handling with Relevant Logic

I was responsible for developing the admin-side backend logic and handling database interactions of the Flood Relief Management System. My main focus was on user management, system reporting, and maintaining database integrity according to the functional requirements of the Admin role.

1. Database Connectivity

- Implemented a reusable MySQL database connection using mysqli_connect
- Added connection error handling using mysqli_connect_error()
- Used a single connection file across backend modules to ensure consistency and maintainability.

2. Admin Backend Functionalities

View Registered Users (Requirement 3.1)

- Developed PHP logic to retrieve all registered users from the user table
- Returned user data in JSON format for frontend rendering
- Enabled fetching detailed information of a selected user for structured display

System Summary & Filtered Reports (Requirement 3.2)

- Designed SQL queries using COUNT() to generate system summaries
- Implemented filtering based on district and relief type
- Generated real-time values such as total users, high severity requests, and requests by selected type

User Deletion with Referential Integrity (Requirement 3.3)

- Identified foreign key constraint issues during user deletion
- Implemented a hybrid cascading approach combining manual deletion and automatic cascading(Deleted related relief requests manually before deleting the user.Used database cascading (ON DELETE CASCADE) to automatically remove related records such as contacts and role-specific data)

3. Admin JavaScript Logic

- Used the Fetch API to communicate with PHP backend endpoints
- Loaded user data dynamically when the admin panel loads
- Updated summary reports without refreshing the page
- Implemented confirmation pop-ups for delete operations to improve user interaction.

Signature:

Name: -U.D.G.I.Erandi

Student ID: -AS20240595

My contribution:

- Created the login page and registration page for the system using HTML, CSS, and JavaScript.
- Make sure users cannot log in without registering first; system gives an alert to register before login.
- Connect the frontend and backend using PHP and MySQL so the system runs fully and smoothly.
- Added role-based access: normal users go to the user dashboard; admins go to the admin panel.
- Implemented role assignment control: only the system owner/admin controller can change a user from normal user to admin.
- Handled and integrated all system components to ensure proper connectivity and smooth operation.
- Identified and resolved errors during development to improve system stability and functionality while maintaining a simple and user-friendly interface. Development issues were addressed using online resources such as Google to identify solutions and fix problems.
- Provided support for final report preparation and documentation.

Signature :